ACTVET
Abu Dhabi Centre for Technical and Vocational Education and Training
مركز أبوظبي للتعليم والتدريب التقني والمهني

مدارس التكنولوجيا التطبيقية
Applied Technology Schools

Computing and Artificial Intelligence Cluster

Capstone Project Proposal



Campus	Baniyas- Girls
Supervisor	MS. Dina Sarris
Capstone Project Title	Shope wise Ai cart
Group Student Names	Dana Ahmed Alya Aref Dan Mohammed Rawda Waleed

1. Project Title and Description

Project Title:

Shop wise AI cart

Project Description:

Shop wise AI cart is an AI-powered mobile app that helps users shop in a smarter, safer, and more budget-friendly way. The system uses the phone camera to scan items directly using AI image recognition. It identifies products, shows details like price and ingredients, and provides real-time alerts such as budget warnings, allergy notifications, and smart suggestions based on user behavior.

2. Problem Statement

Clearly describe the problem your project aims to solve.Explain why this problem is important.

2. Problem Statement

Many people shop without planning due to impulsive or emotional decisions, which leads to .overspending, buying unnecessary items, and sometimes choosing products that may cause allergies

.This problem is important because it affects financial control, health, and daily decision-making

3. Background Research

Summarize existing solutions or research related to your project.Explain gaps your project will address.

Some existing solutions include image recognition apps, shopping assistants, and budget tracking tools. However, these systems are separate and do not provide real-time support during shopping.

Shop wise AI cart solves this by combining AI image recognition, behavior analysis, and safety features in one system.

4. Project Scope

Define what your project will include and what it will NOT include.Mention limitations if any.

Includes:

- Mobile app with camera-based item scanning
- AI image recognition
- Budget tracking and alerts
- Allergy detection
- Behavior analysis (impulsive/repeated buying)
- Real-time suggestions

Does NOT include:

- Full supermarket database
- Advanced machine learning models
- Physical cart hardware

Limitations:

- Depends on image recognition accuracy
- Limited dataset

5. Platforms, AI Tools, Software and Resources

List all tools and technologies used:- Programming languages- AI tools/libraries- Software- Hardware (if any)- Datasets

- Programming languages: Dart / JavaScript
- Framework: Flutter or React Native
- AI tools: Image recognition APIs
- Software: Visual Studio Code
- Database: Firebase
- Hardware: Mobile phone camera
- Dataset: Manually created product data

6. Expected Outcomes

Explain what results your project will achieve. Mention deliverables such as system, prototype, or report.

- A working mobile app prototype Shop wise AI cart
- Item recognition using camera
- Real-time alerts (budget, allergy, suggestions)
- AI-based behavior analysis
- Final report and presentation